

A Report on
PWM Generator in Verilog Using Xilinx On FPGA
for
Mini Project 2B (REV- 2019 'C' Scheme) of Third Year, (TE Sem-VI)
in
Electronics & Telecommunication Engineering

by

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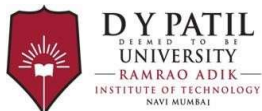
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UNIVERSITY OF MUMBAI

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CERTIFICATE

This is to certify that the project entitled **PWM Generator in Verilog Using Xilinx On FPGA** is a bonafide work of

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submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Mini Project 2B (REV- 2019 'C' Scheme) of Third Year, (TE Sem-VI) in Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2021-22**

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1.INTRODUCTION

1.1 Need :

1.Nowadays technology has become an important part of our world helping to complete and automate tasks in less time with no human assistance required.

2.Home Equipment use FPGA Boards very commonly to make lives easier.

3.PWM is very useful in Devices using Light Emitting Diode, to Control their Intensity.

1.2 Definition:

Pulse-width modulation (PWM), or **pulse-duration modulation (PDM)**, is a method of reducing the average power delivered by an electrical signal, by effectively chopping it up into discrete parts. The average value of voltage (and current) fed to the load is controlled by turning the switch between supply and load on and off at a fast rate.

The longer the switch is on compared to the off periods, the higher the total power supplied to the load. Along with maximum power point tracking (MPPT), it is one of the primary methods of reducing the output of solar panels to that which can be utilized by a battery.

PWM is particularly suited for running inertial loads such as motors, which are not as easily affected by this discrete switching, because their inertia causes them to react slowly. The PWM switching frequency has to be high enough not to affect the load, which is to say that the resultant waveform perceived by the load must be as smooth as possible.

The rate (or frequency) at which the power supply must switch can vary greatly depending on load and application. For example, switching has to be done several times a minute in an electric stove; 100 or 120 Hz (double of the utility frequency) in a lamp dimmer; between a few kilohertz (kHz) and tens of kHz for a motor drive; and well into the tens or hundreds of kHz in audio amplifiers and computer power supplies.

The main advantage of PWM is that power loss in the switching devices is very low. When a switch is off there is practically no current, and when it is on and power is being transferred to the load, there is almost no voltage drop across the switch.

Power loss, being the product of voltage and current, is thus in both cases close to zero. PWM also works well with digital controls, which, because of their on/off nature, can easily set the needed duty cycle. PWM has also been used in certain communication systems where its duty cycle has been used to convey information over a communications channel.

In electronics, many modern microcontrollers (MCUs) integrate **PWM controllers** exposed to external pins as peripheral devices under firmware control by means of internal programming interfaces. These are commonly used for direct current (DC) motor control in robotics and other applications.

PWM works by pulsating DC current, and varying the amount of time that each pulse stays 'on' to control the amount of current that flows to a device such as an LED. PWM is digital, which means that it has two states: on and off (which correspond to 1 and 0 in the binary context, which will become more relevant to you if using microcontrollers).

The longer each pulse is on, the brighter the LED will be. Due to the fact that the interval between pulses is so brief, the LED doesn't actually turn off. In other words, the LED's power source switches on and off so fast (thousands of times per second) that the LED actually stays on without flickering. This is called PWM dimming, and such a circuit is just called a PWM LED dimmer circuit.

The squares in the PWM illustration below are the pulses which represent 'on' time, and the depressed areas represent the time that the power is 'off'. Both the squares and depressed areas are the same 'width', therefore the duty cycle is 50%. PWM signals are typically square waves.

If the duty cycle of a PWM power supply is set to 70%, then the pulse is on for 70% of the time, and it is off 30% of the time. Duty cycle refers to the amount of time it is on. At a 70% duty cycle, an LED's brightness should be near 70%. The correlation between duty cycle and brightness is not 100% linear, as the efficiency of LEDs varies with the amount of current supplied.

If the duty cycle was 0%, the entire signal would be flat, as shown below. A PWM duty cycle of 0% means that the power is off. In such a state, an LED would not be operational. It would simply be off.

2.COMPARATIVE STUDY (Similar projects done previously)

1.In 21st century, as human beings are becoming more productive in their desired work; they are trying to reduce the time required for the unwanted daily activities by introducing different concepts in FPGA. In a vast country like India, it is important for us to be susceptible and adaptive .

2.After understanding this we have tried to learn different method and uses of PWM using Verilog and applying them on Xilinx Simulator and then FPGA board.

3.The first project which we took to understand the basics of PWM was to consider only one duty cycle of 50% applied to control the intensity of only a single LED.

4.Moving towards more complex concepts we designed controlling multiple LED's simultaneously

5.Now, after clearing our basics and realizing the FPGA we have decided to explore a project which can be useful for solving arithmetic modulation problems using FPGA.

3.PROBLEM STATEMENT

The major uses of LED (Light Emitting Diodes) are to illuminate objects and even places. Its application is everywhere due to its compact size, low consumption of energy, extended lifetime, and flexibility in terms of use in various applications.

Few LED applications include dimming of lights which helps in reducing energy consumption.

This dimming feature is also used in Appliances where it is of two types.

1.Global Dimming where all LEDs are dimmed together.

2.Local Dimming where LEDs are dimmed independently.

Dimming of LED lights is one of the most common difficulties when converting to LED lighting. However, this is primarily due to the fact that many people simply assume that LED's can be dimmed just as easily as the old light sources. If this does not work without problems, sometimes the entire LED technology is demonized as not mature.

The brightness of light emitting diodes is not regulated by the voltage, but by the current or a so-called pulse width modulation. This regulation as well as the conversion of the mains voltage into the internal low voltage is done by a control electronics integrated in the LED lamp. If the LED lamp should be dimmed, this internal circuit must be designed for this purpose.

Our project demonstrates how a simple and fast pulse width modulator (PWM) generator can be implemented using Verilog programming. It is simulated using Xilinx ISE, a multi-language (hardware description language) simulation environment from Xilinx and tested on Elbert V2 development board from Numato Lab.

4.MINI PROJECT DESIGN (PRINCIPLE AND WORKING)

4.1 Block Diagram

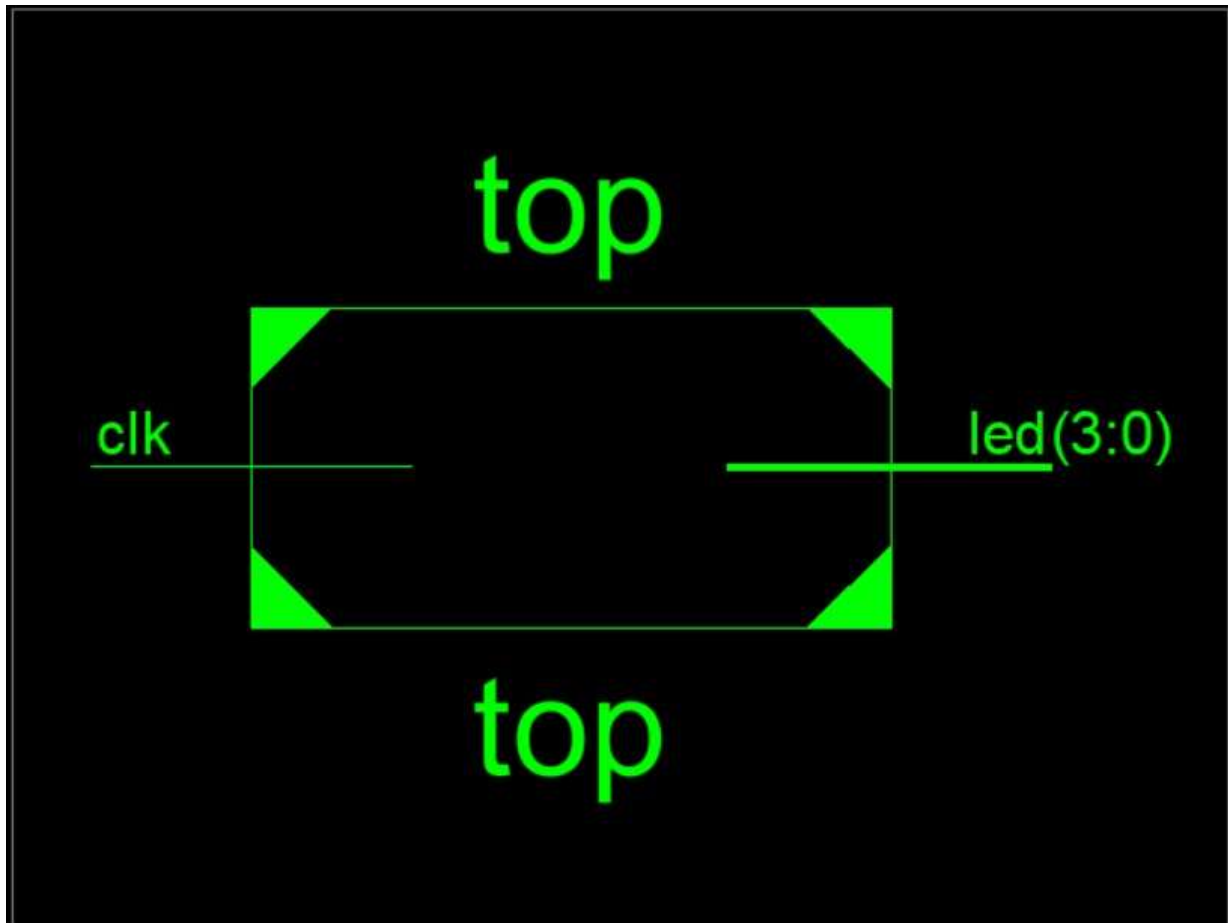


Figure 1. Block diagram generated in ISE

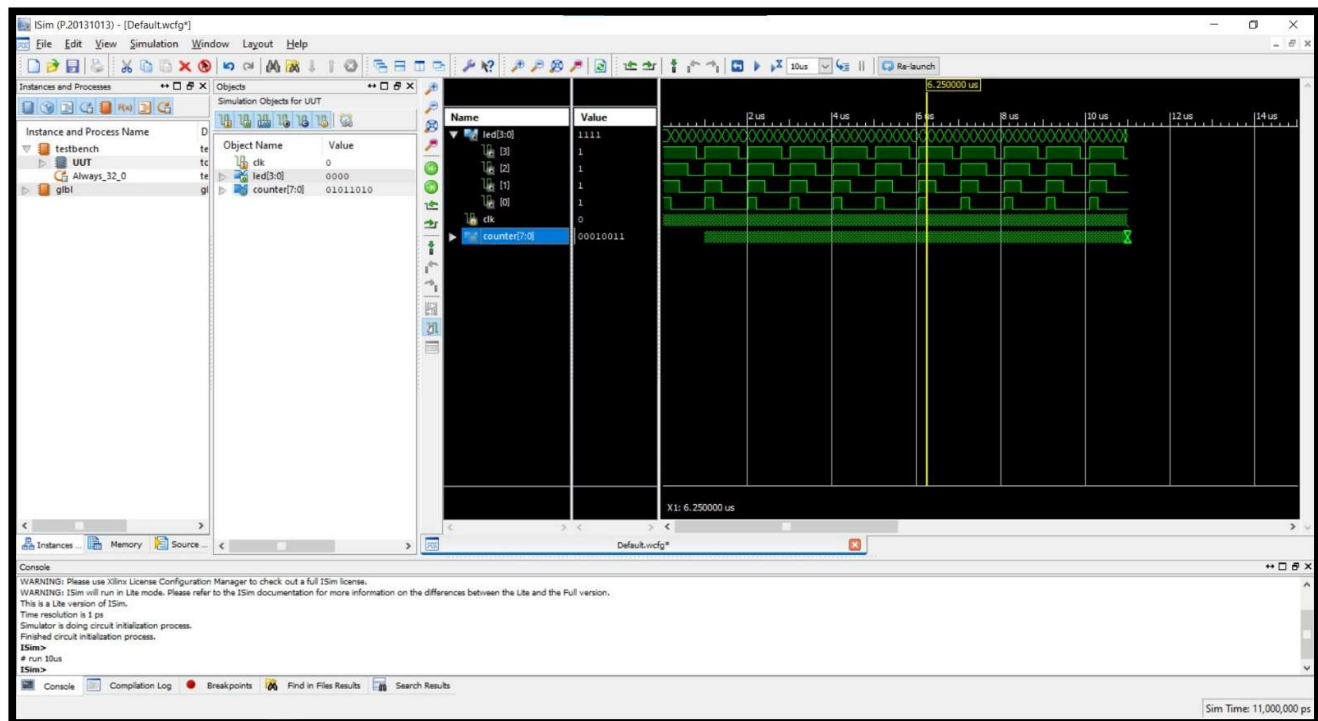


Figure 2. Simulation and generation of PWM output for four LEDs successfully.

4.2 Block Diagram Description:

Here in the block diagram we can see that we used the one clock operand, eight counters and four LEDs.

4.3 Circuit diagram and Working

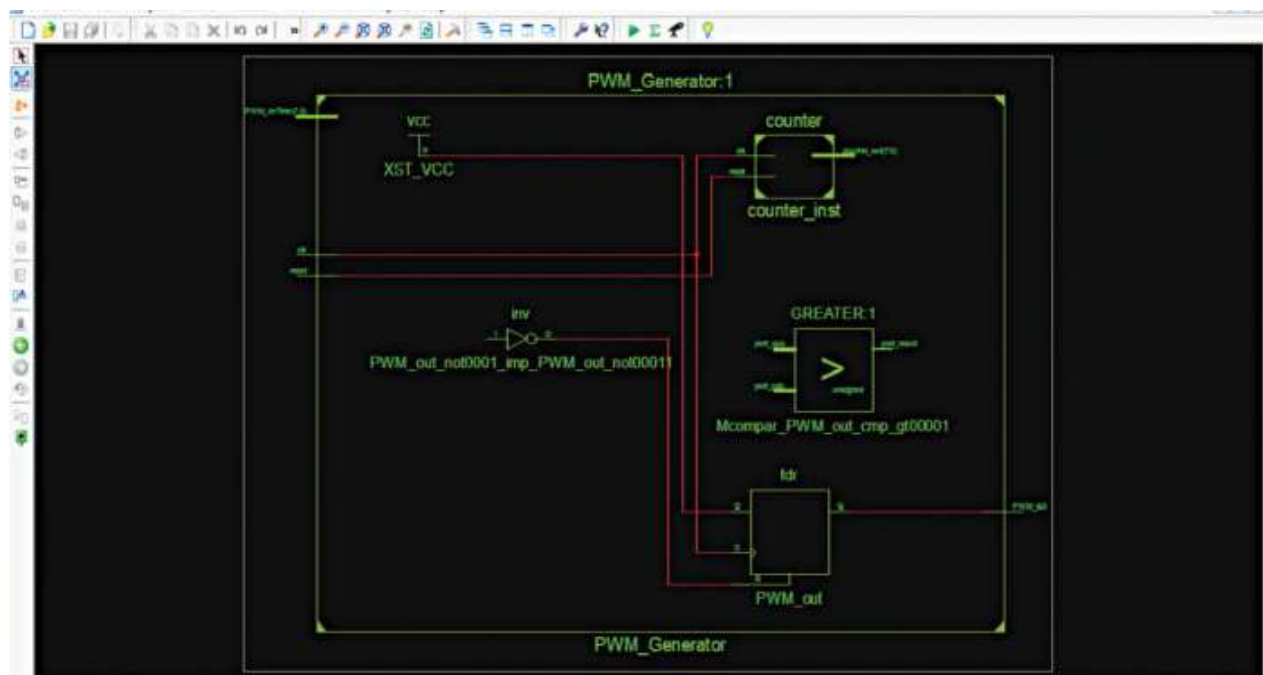


Figure 3. Circuit diagram simulation for PWM generator in Xilinx ISE.

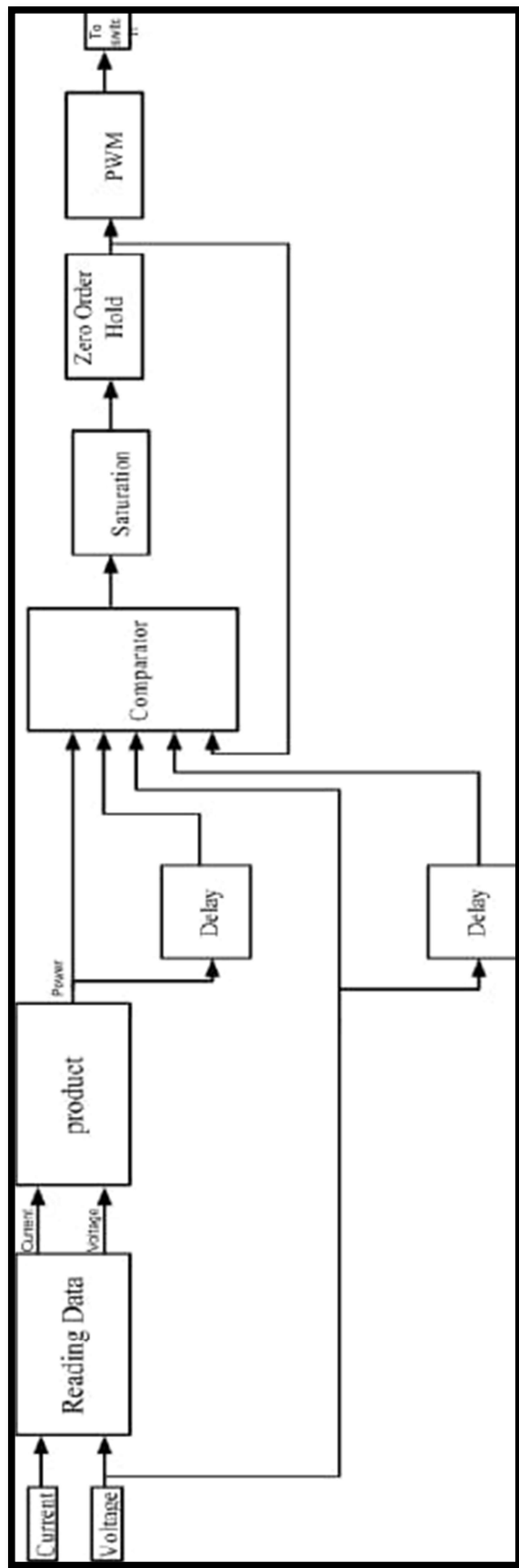


Figure 4. Block circuit diagram of PWM generator.

Working:

Block diagram of the PWM generator is shown. Working principle of the generator is simple. It uses one counter and one comparator. The microcontroller unit provides 8-bit input into PWM module. Counter used in the PWM module is 8-bit. It increments its value on the positive edge of the clock (positive edge triggered). Comparator used in the PWM generator is also 8-bit.

Input given to PWM module is compared to the current value of the counter using the comparator. If current value of the counter is greater than the value given to the module as input, PWM output is pulled low. However, if current value of the counter is less than the value given as input to the module, output of PWM generator is pulled high.

Let us take a test case to understand the operation. Suppose, input to PWM module is 128 (10000000_b). Counter is initialised using Reset button, so that output is 0 (00000000_b). During the first clock cycle, value of counter and input value to PWM module are compared.

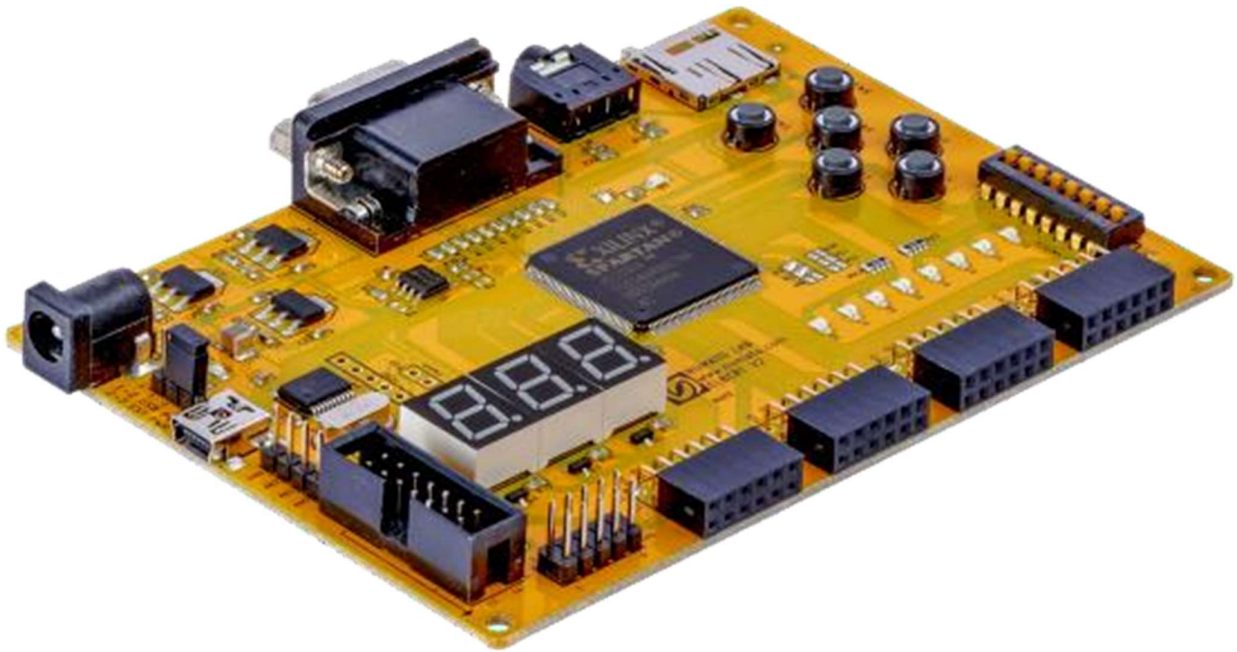
The module finds that, value of counter is less than the value provided. This prompts the module to pull output of PWM generator high. This same paradigm keeps output of the generator pulled high until value of the counter reaches 128 (10000000_b).

In the next clock cycle, after the counter has reached 128, counter increments its value to 129 (10000001_b). Comparator finds that, current counter value is greater than the value provided to PWM module, therefore output must be pulled low.

Changing input to PWM module will consequently change the threshold value, where transition from high to low state occurs. Thus, PWM can be achieved by varying the input provided to PWM module.

5.COMPONENTS/TOOL TO BE USED

5.1 Components (with features related to project)



ELBERT V2 is a simple but versatile FPGA Learning/Development board featuring Xilinx Spartan 3A FPGA. An excellent choice for beginners and advanced learners for experimenting and learning system design with FPGAs. This development board features Xilinx XC3S50A 144 pin FPGA with a maximum of 108 user I/Os (Some I/Os are dedicated to system and peripherals). USB interface provides a fast and easy configuration download to the onboard SPI flash. You don't need a programmer or special downloader cable to download bitstream to Elbert. ELBERT V2 features a stable clock source that is derived from the onboard configuration controller. ELBERTV2 incorporates LEDs, switches, and other peripherals for curious users to get started with the "Hello World" program in a matter of minutes.

Features

- FPGA: Spartan XC3S50A in TQG144 package
- Flash memory: 16 Mb SPI flash memory (M25P16)
- USB 2.0 interface for On-board flash programming
- FPGA configuration via JTAG and USB
- 8 LEDs, Six Push Buttons and 8 way DIP switch for user-defined application
- VGA output
- Stereo audio out
- Micro SD Card Adapter
- Three Seven Segment Displays
- 39 IOs for user-defined purposes
- On-board voltage regulators for single power rail operation
- RoHS Compliant

Applications

- Product Prototype Development
- Home Networking
- Signal Processing
- Wired and Wireless Communications
- Educational tool for schools and universities

5.2 Software used:



ISE:-

ISE primarily targets development of embedded firmware for Xilinx FPGA and CPLD integrated circuit (IC) product families. It was succeeded by Xilinx Vivado. Use of the last released edition from October 2013 continues for in-system programming of legacy hardware designs containing older FPGAs and CPLDs otherwise orphaned by the replacement design tool, Vivado Design Suite.

ISE enables the developer to synthesize ("compile") their designs, perform timing analysis, examine RTL diagrams, simulate a design's reaction to different stimuli, and configure the target device with the programmer. Other components shipped with the Xilinx ISE include the Embedded Development Kit (EDK), a Software Development Kit (SDK) and ChipScope Pro. The Xilinx ISE is primarily used for circuit synthesis and design, while ISIM or the ModelSim logic simulator is used for system-level testing.

Xilinx ISE is tightly-coupled to the architecture of Xilinx's own chips (the internals of which are highly proprietary) and cannot be used with FPGA products from other vendors. Given the highly proprietary nature of the Xilinx hardware product lines, it is rarely possible to use open source alternatives to tooling provided directly from Xilinx, although as of 2020, some exploratory attempts are being made.

System-level testing may be performed with ISIM or the ModelSim logic simulator, and such test programs must also be written in HDL languages. Test bench programs may include simulated input signal waveforms, or monitors which observe and verify the outputs of the device under test.

ModelSim or ISIM may be used to perform the following types of simulations:

- Logical verification, to ensure the module produces expected results
- Behavioural verification, to verify logical and timing issues
- Post-place & route simulation, to verify behaviour after placement of the module within the reconfigurable logic of the FPGA

The ISE Design Suite: Embedded Edition includes Xilinx Platform Studio (XPS), Software Development Kit (SDK), large repository of plug and play IP including MicroBlaze™ Soft Processor and peripherals, and a complete RTL to bit stream design flow.

Embedded Edition provides the fundamental tools, technologies and familiar design flow to achieve optimal design results. These include intelligent clock gating for dynamic power reduction, team design for multi-site design teams, design preservation for timing repeatability, and a partial reconfiguration option for greater system flexibility, size, power, and cost reduction.

Xilinx:

Xilinx ISE (Integrated Synthesis Environment) is a discontinued software tool from Xilinx for synthesis and analysis of HDL designs, which primarily targets development of embedded firmware for Xilinx FPGA and CPLD integrated circuit(IC) product families.

It was succeeded by Xilinx Vivado. Use of the last released edition from October 2013 continues for in-system programming of legacy hardware designs containing older FPGAs and CPLDs otherwise orphaned by the replacement design tool, Vivado Design Suite.

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As commonly practiced in the commercial electronic design automation sector, Xilinx ISE is tightly-coupled to the architecture of Xilinx's own chips (the internals of which are highly proprietary) and cannot be used with FPGA products from other vendors.

The Spartan series targets low cost, high-volume applications with a low-power footprint e.g. displays, set-top boxes, wireless routers and other applications.

The Spartan-6 family is built on a 45 nm, 9-metal layer, dual-oxide process technology. The Spartan-6 was marketed in 2009 as a low-cost option for automotive, wireless communications, flat-panel display and video surveillance applications.

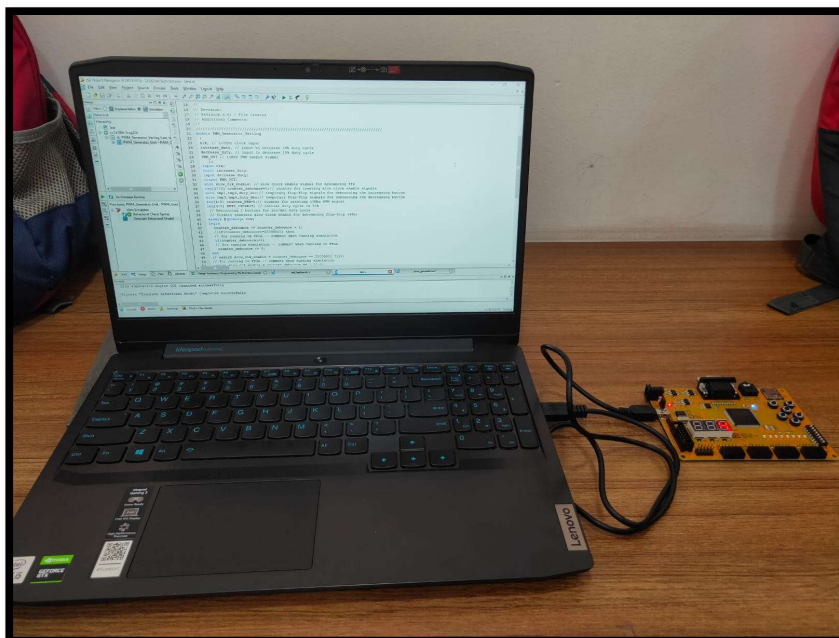
The Spartan-7 family, built on the same 28 nm process used in the other 7-Series FPGAs, was announced in 2015, and became available in 2017. Unlike the Artix-7 family and the "LXT" members of the Spartan-6 family, the Spartan-7 FPGAs lack high-bandwidth transceivers

6.PROPOSED EXECUTIONS STEPS

6.1 Implemented Steps for PCB

1. Here we doing the implementation by using the FPGA BOARD hardware for PWM generator.
2. We connected the all the required connections from FPGA board to computer using a USB cable.
3. We made the connection to the burn the code on the FPGA board form the computer system to give us the desired output.

6.2 FPGA board layout and Verilog code (actual photos of project from front and back view)



7.TROUBLESHOOTING

5.1 Problems/Faults in project

1.NO SIGNIFICANT ERROR/ PROBLEMS ENCOUNTERED..

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